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1. Introduction

1.1 About the Project

Laptop Pasal is a modern e-commerce platform developed to provide customers with a convenient and reliable way to purchase a wide range of laptops online. The name Laptop Pasal reflects the concept of a digital marketplace (“Pasal” meaning shop), where users can explore, compare, and buy laptops from the comfort of their homes.

The main objective of Laptop Pasal is to deliver a smooth and user-friendly online shopping experience for customers looking for laptops suitable for various needs, such as education, professional work, gaming, or personal use. The platform allows users to browse products through organized categories, view detailed product information, add items to a shopping cart, and complete purchases efficiently. Features such as user registration, login, and order tracking further enhance the overall shopping experience.

The website is developed using WordPress and the WooCommerce plugin, ensuring flexibility, scalability, and ease of management. It is hosted locally using XAMPP for development purposes, and integrates the Cash on Delivery system to provide secure and trustworthy online transactions. Overall, Laptop Pasal demonstrates the effective use of modern web technologies to build a functional, secure, and visually appealing e-commerce solution.

1.2 Problem Statement

- Customers often find it difficult to locate a reliable online platform that focuses mainly on laptops and daily tech items.
- Many e-commerce websites lack clear and well-organized product categories, making product browsing time-consuming.
- Insufficient or unclear product descriptions and specifications can prevent customers from making informed purchase decisions.
- Complicated navigation and an unfriendly user interface may lead to a poor shopping experience and cart abandonment.
- Concerns about secure online payments reduce customer trust in online technology stores.

Laptop Pasal aims to overcome these challenges by providing a focused, well-structured, and secure e-commerce platform dedicated to laptops and essential tech products, ensuring a smooth and trustworthy online shopping experience.

1.3 Objectives

The main objectives of the **Laptop Pasal** project are as follows:

1. **User-Friendly Interface:** To develop an easy-to-navigate website that allows users to quickly browse and search for laptops and daily tech items based on categories, brands, and specifications.
2. **Detailed Product Information:** To provide clear and comprehensive product details, including specifications, features, and pricing, helping customers make informed purchasing decisions.
3. **Secure Order Processing:** To integrate Cash On Delivery as a reliable payment gateway, ensuring safe and secure online transactions for users.
4. **Responsive and Mobile-Friendly Design:** To ensure the website is fully responsive and accessible across desktops, tablets, and mobile devices, providing a consistent and smooth shopping experience on all platforms.

1.4 Proposed System

The **Laptop Pasal** e-commerce platform is developed using **WordPress** and powered by the **WooCommerce** plugin, providing the flexibility and scalability required to manage an online technology store. The system offers the following key features:

- **Product Catalog:** Laptops and daily tech items are organized into clear categories such as brands, usage types, and accessories. Each product includes a detailed product page with images, pricing, and technical specifications.
- **Customer Accounts:** Users can create personal accounts to manage their profiles, track orders, save shipping details for faster checkout, and view their purchase history.
- **Secure Checkout:** The checkout process is simple and reliable, with **Cash On Delivery** integrated to ensure secure and trustworthy checkouts.
- **Mobile Compatibility:** The website is fully responsive and optimized for both desktop and mobile devices, allowing users to browse and shop anytime and anywhere.

The Laptop Pasal platform aims to deliver a convenient, efficient, and secure online shopping experience for customers looking to purchase laptops and essential technology products.

2. Requirements Analysis

2.1 Functional Requirements

The functional requirements define the core functionalities and features of the e-commerce website **Laptop Pasal**. These include:

1. **User Registration and Login:** Customers can create accounts, log in, and manage their profiles.
2. **Product Catalog:** The website should display a categorized list of laptops and accessories.
3. **Product Details:** Each product should have a dedicated page showing detailed information like description, price, and images.
4. **Shopping Cart:** Customers can add products to their shopping cart, view items, update quantities, or remove products.
5. **Order Placement:** Customers can place orders by providing necessary details (shipping address,).
6. **Order Management:** Admin can view, update, or cancel orders.
7. **Payment Integration:** The website implements the cash on delivery payment system.
8. **Search and Filter:** Customers can search for specific products and filter based on category or price.

2.2 Non-Functional Requirements

Non-functional requirements focus on system performance and user experience. These include:

1. **Usability:** The website should be intuitive and user-friendly, with easy navigation for both customers and admins.
2. **Performance:** The system should ensure fast load times (under 3 seconds) even with a moderate number of concurrent users.
3. **Scalability:** The website should be able to handle an increasing number of products, users, and orders as the business grows.
4. **Reliability:** The system must be available 24/7 with minimal downtime, ensuring smooth operations.
5. **Security:** All customer data, including personal information and payment details, should be protected through secure connections (HTTPS) and payment gateway encryption.
6. **Compatibility:** The website should work seamlessly across different devices and browsers (Chrome, Firefox, Safari, mobile devices).

3. System Requirements

3.1 Hardware Requirements

For the local setup and deployment of the **Laptop Pasal** website, the following hardware requirements are recommended:

- **Web Server:** A computer or server with a minimum of 4GB RAM and 50GB of free disk space to host the website, images, and the database.
- **Processor:** Intel Core i5 or equivalent, with at least 2 cores, to ensure smooth operation of the XAMPP local server environment.
- **Backup Storage:** External or cloud-based storage for regular backups of product data, images, and user orders.

3.2 Software & Tools Used

The following software and tools were utilized to build and deploy the **Laptop Pasal** website, which sells laptops and other products:

Software/Tool	Purpose
XAMPP	Provides a local server environment with Apache, MySQL, and PHP. Used to host and run WordPress and WooCommerce locally during development and testing.
WordPress	Content management system (CMS) for building and managing the Laptop Pasal website, including product pages, content, and administrative functions.
WooCommerce	E-commerce plugin for WordPress that enables product listings, shopping cart, checkout, and order management features.
Payment Integration	Payment gateway integrated into WooCommerce for securely processing cash on delivery purchases.
PHP	Server-side scripting language used for dynamic content and backend management on the website.
MySQL	Database management system, included with XAMPP, used for storing product details, user data, and orders.

Web Browser	For testing the website and ensuring compatibility across different browsers (e.g., Google Chrome, Firefox, Safari).
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4. System Design

This section outlines how the Laptop Pasal e-commerce website is structured and how users interact with it. It explains the flow of data, the relationship between key components, and the various functions performed by users such as browsing laptops, placing orders, and making payments. Visual diagrams are used to represent the internal structure and external interactions of the system, helping to understand how the website operates both from a user and backend perspective.

4.1 Entity- Relationship Diagram(ERD):

Describes how data is organized in the system. It includes main entities like Users,Products, Orders, and Payments, and shows how they are linked—for example, a user can include placing multiple orders, and each order includes one or more products.

E-commerce Order and Payment Data Model

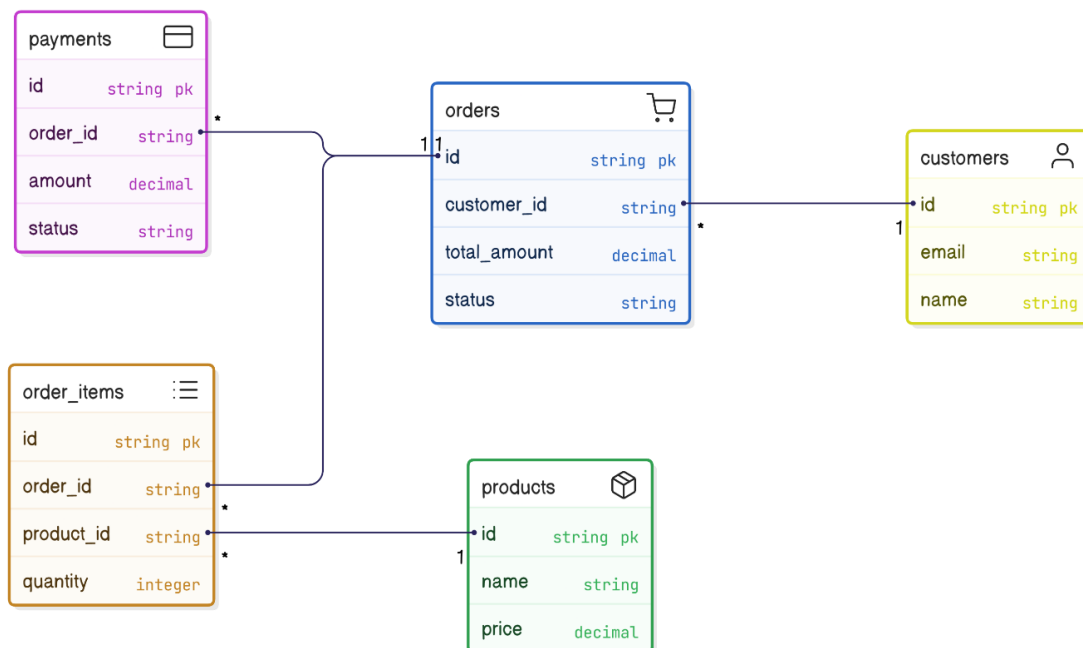


Fig: ERD Diagram

4.2 Use Case Diagram:

Illustrates the actions that users (customers and admin) can perform in the system. Customers can browse, register, place orders, and make payments, while the admin can manage products, view orders, and update inventory.

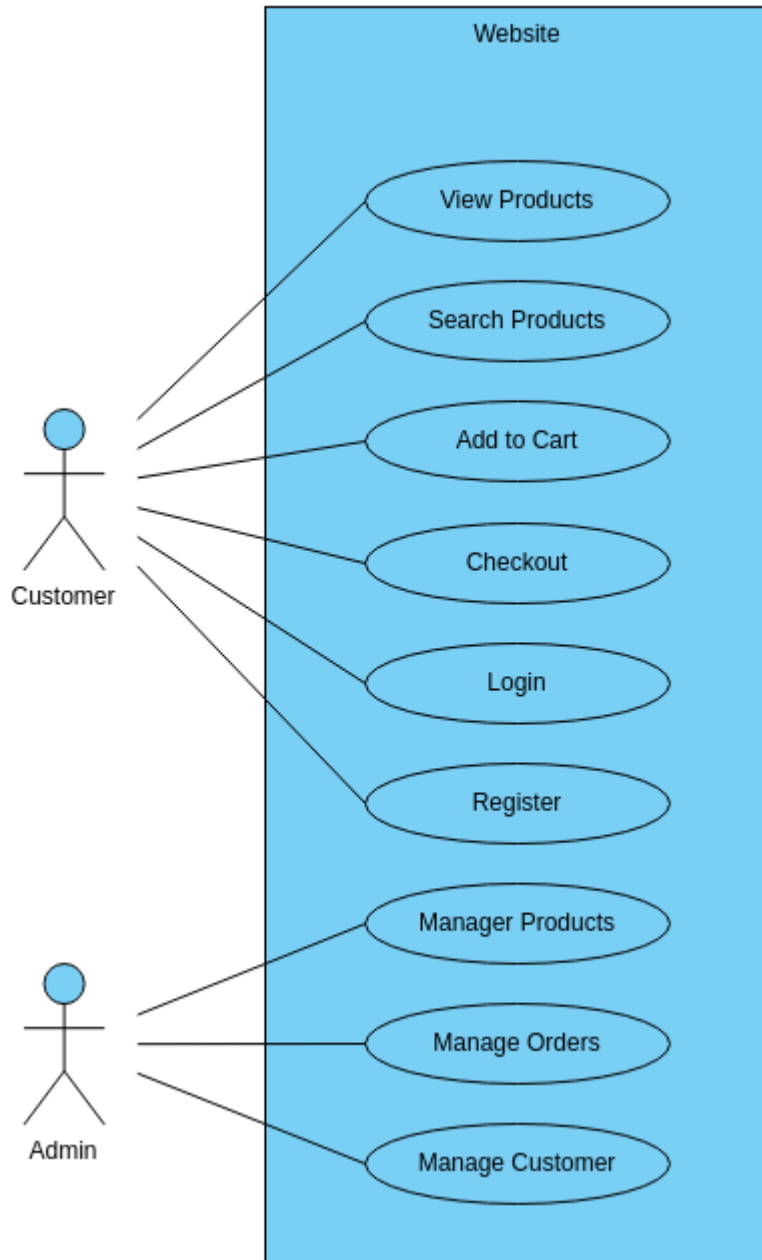


Fig:Use Case Diagram

4.Implementation

The platform was built using WordPress and the WooCommerce plugin, leveraging their functionalities to meet the project's objectives. Key features were implemented to ensure a user-friendly shopping experience, secure transactions, and effective product presentation. The development was carried out locally using the XAMPP environment.

5.1 Key Features

The core functionalities of the Laptop Pasal website were implemented as follows:

5.1.1 Product Catalog

The product catalog was created using WooCommerce's product management features. products were added as individual products, categorized into "Laptops, accessories and Gaming" for easier navigation. Each product entry includes:

- A descriptive title (e.g., "Apple MacBook M1," "Acer Nitro 5").
- High-quality images showcasing the laptops and other products.
- A detailed description covering characteristics and basic care tips, fulfilling the objective of providing comprehensive information.
- The price of the product.
- Relevant categories and tags to aid searching and filtering.

This setup allows customers to browse the available selection easily and access necessary details directly on the product page.

5.1.2 Shopping Cart

The shopping cart functionality is powered by WooCommerce's built-in system. Users can:

- Add desired products to their cart directly from the product or shop pages using an "Add to Cart" button.
- View the contents of their cart at any time via the cart icon/page.
- Adjust the quantity of items or remove items from the cart.
- See a subtotal of their selected items before proceeding to checkout.

This provides a standard and intuitive e-commerce cart experience.

5.1.3 Cash on Delivery Integration

To provide customers with a flexible and convenient payment option, **Cash on Delivery (COD)** was integrated into the WooCommerce platform. This was achieved by:

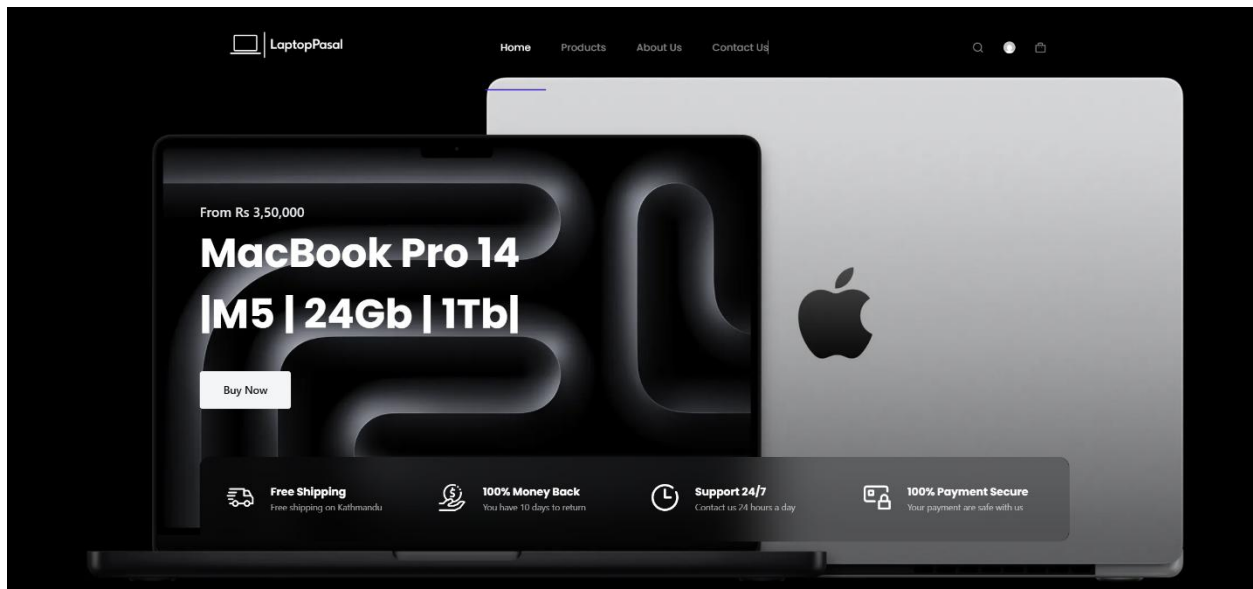
- Enabling the built-in **Cash on Delivery** payment method available in WooCommerce.
- Configuring the COD settings to restrict availability based on delivery locations and shipping methods, ensuring operational feasibility.
- Customizing the option label and instructions to clearly inform customers that payment will be collected upon product delivery.

During the checkout process, users selecting **Cash on Delivery** can place their order without making an online payment. The order is confirmed on the Laptop Pasa! website, and payment is collected directly from the customer at the time of delivery. This integration fulfills the objective of offering an accessible and user-friendly payment alternative for customers who prefer offline payment methods.

5.2 Screenshots of Web Pages

The following subsections provide visual examples of the implemented Laptop Pasa! website's user interface, showcasing the key pages involved in the customer journey.

5.2.1 Homepage



Laptops

3



Accessories

4



Gaming

4



Nothing Headphone (1)

Come to Play

[View Offer](#)



Featured Products

[View All](#)



Asus ROG Zephyrus G14
[16GB RAM|1TB SSD]
[Ryzen 9 8945HS|RTX 4060]

Rs 289,000.00

GAMING, LAPTOPS

[Add to cart](#)



Lenovo Slim 7 Pro X
[Ryzen 9 6900HS|RTX 3050]
[32GB DDR5 RAM|1TB SSD]

Rs 170,000.00

LAPTOPS

[Add to cart](#)



Logitech Astro A10

Rs 20,999.00

ACCESSORY, GAMING, HEADPHONES

[Add to cart](#)

5.2.2 Product Page

 LaptopPasal

[Home](#) [Products](#) [About Us](#) [Contact Us](#)

HOME / PRODUCTS

Products

SHOWING ALL 7 RESULTS

Default sorting



Asus ROG Zephyrus G14
[16GB RAM][1TB SSD]
[Ryzen 9 8945HS][RTX 4060]
Rs 289,000.00
GAMING, LAPTOPS
Add to cart



Lenovo Slim 7 Pro X
[Ryzen 9 6900HS][RTX 3050]
[32GB DDR5 RAM][1TB SSD]
Rs 170,000.00
LAPTOPS
Add to cart



Logitech Astro A10
Rs 20,999.00
ACCESSORY, GAMING, HEADPHONES
Add to cart



Logitech Pro 2 Lightspeed Wireless Mouse
Rs 12,000.00
ACCESSORY, GAMING
Add to cart



MacBook Pro 14
[M5 | 24Gb Ram | 1Tb SSD]
Rs 350,000.00
LAPTOPS
Add to cart

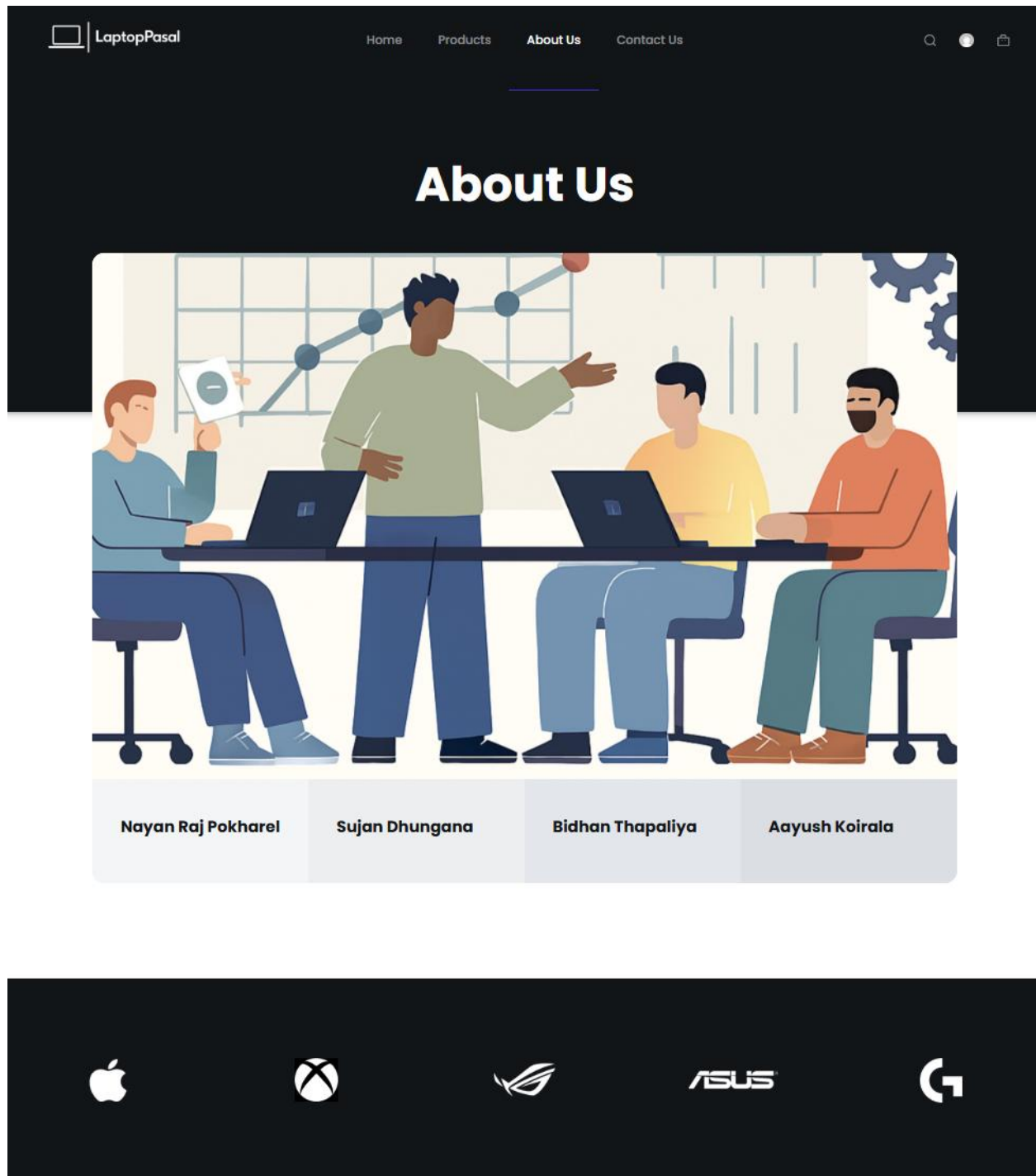


Nothing Headphones (1)
Rs 35,000.00
ACCESSORY, HEADPHONES
Add to cart



XBOX Elite Controller 2
Rs 20,000.00
ACCESSORY, GAMING
Add to cart

5.2.3 About Us



Testimonials



Absolutely impressed with the service! The laptop arrived earlier than expected, perfectly packaged, and exactly as described. Their customer support was responsive and helped me choose the right model for my needs. Highly recommended for anyone looking for reliable tech online!

Yugal Poudel



This store is a hidden gem! I got a great deal on a high-performance laptop, and the checkout process was smooth and secure. The product was genuine, and the warranty was properly registered. Will definitely shop here again!

Famous Dhungana

5.2.4 Checkout Page

 LaptopPasal

[Home](#) [Products](#) [About Us](#) [Contact Us](#)

Cart

Product	Quantity	Subtotal
 Asus ROG Zephyrus G14 [16GB RAM][1TB SSD] [Ryzen 9 8945HS][RTX 4060] Rs 289,000.00	- 1 +	Rs 289,000.00 
 Logitech Pro 2 Lightspeed Wireless Mouse Rs 12,000.00	- 1 +	Rs 12,000.00 

Coupon code

Apply coupon

Update cart

Cart totals

Subtotal Rs 301,000.00

Shipping

Free shipping

Shipping to **Bagmati**.

Change address

Total **Rs 301,000.00**

Proceed to checkout

Checkout

Billing details

First name *

Last name *

Company name (optional)

Country / Region *

Street address *

Town / City *

State / Zone *

Postcode / ZIP (optional)

Phone *

Email address *

☐ I would like to receive exclusive emails with discounts and product information

Your order

Product	Subtotal
Asus ROG Zephyrus G14 [16GB RAM][1TB SSD] [Ryzen 9 8945HS][RTX 4060] × 1	Rs 289,000.00
Logitech Pro 2 Lightspeed Wireless Mouse × 1	Rs 12,000.00
Subtotal	Rs 301,000.00
Shipping	
Free shipping	
Total	Rs 301,000.00

☒ Cash on delivery

Pay with cash upon delivery.

Your personal data will be used to process your order, support your experience throughout this website, and for other purposes described in our [privacy policy](#).

Place order

Checkout

 Thank you. Your order has been received.

Order number: 1175	Date: December 20, 2025
Total: Rs 301,000.00	Payment method: Cash on delivery

Pay with cash upon delivery.

Order details

Product	Total
Asus ROG Zephyrus G14 [16GB RAM][1TB SSD] [Ryzen 9 8945HS][RTX 4060] × 1	Rs 289,000.00
Logitech Pro 2 Lightspeed Wireless Mouse × 1	Rs 12,000.00
Subtotal:	Rs 301,000.00
Shipping:	Free shipping
Total:	Rs 301,000.00
Payment method:	Cash on delivery

Billing address

Nayan Pokharel
BANESWOR
Kathmandu
Bagmati
+9779862354666

pokharelnayan03@gmail.com

Shipping address

Nayan Pokharel
BANESWOR
Kathmandu
Bagmati

5.2.5 Login Page

Username or Email Address

Nayan

Password

••••••••



☐ Remember Me

[Forgot Password?](#)

Log In

6. Testing & SEO

This section includes basic testing activities performed to ensure the core functionalities of the website work as expected, along with basic SEO strategies applied to improve visibility on search engines.

6.1 Basic Testing

Test Case	Expected Result	Status
User Registration	New user can register with valid details	Passed
User Login	Registered user can log in successfully	Passed
Browse Products	Products are displayed correctly on the homepage	Passed
Add to Cart	Selected products are added to the cart	Passed
Remove from Cart	Products can be removed from the cart	Passed
Checkout Process	User can proceed to checkout	Passed
Admin Product Management	Admin can add/edit/delete products	Passed
Responsive Design Test	Website works well on mobile and desktop	Passed

6.2 SEO Strategies

- **Meta Tags:** Added relevant title and description tags to web pages.
- **Keyword Optimization:** Used laptop and its accessories related keywords (e.g., budget laptop, gaming laptop, etc) in content.
- **Image Alt Text:** Provided alt tags for product images to boost search visibility.
- **Mobile Friendliness:** Ensured responsive design for better SEO ranking.
- **Readable URLs:** Used SEO-friendly URLs through WordPress permalink settings.

7. Conclusion and Future Scope

This section outlines the key accomplishments of the Laptop Pasal project, acknowledges its current limitations, and presents suggestions for future enhancements.

7.1 Achievements

- Developed a fully functional e-commerce website for selling laptops and related accessories.
- Utilized WordPress along with WooCommerce for efficient product and order management.
- Integrated checkout system to enable secure online payment transactions.
- Ensured basic mobile responsiveness and implemented fundamental SEO strategies.
- Designed a clean and user-friendly interface to enhance customer experience.

7.2 Limitations

- Website is currently running locally through XAMPP and is not publicly hosted.
- Only Cash on Delivery is available as a payment option, limiting accessibility for some users.
- Features such as inventory tracking, delivery system, and user reviews are not yet included.
- The admin dashboard lacks advanced analytics and reporting functionalities.

7.3 Future Improvements

In order to further improve the Laptop Pasal platform, the following enhancements are suggested:

- It is recommended to host the website on a live server, allowing real-time public access and increasing its availability to a wider audience.
- To provide more convenience for local users as well as international users, it would be beneficial to integrate additional payment gateways such as eSewa, Khalti and Paypal.
- Adding features like order tracking, customer reviews, and informative blogs will help enhance user engagement and offer added value to customers.
- Strengthening the admin panel with features such as detailed analytics, inventory control, and sales reports will help streamline operations and improve decision-making.

8. References

1. **WordPress.org.** *WordPress.* <https://wordpress.org>
2. **WooCommerce.** *WooCommerce: The most customizable eCommerce platform for building your online business.* <https://woocommerce.com>
3. **Apache Friends.** *XAMPP for Windows.* <https://www.apachefriends.org/index.html>
4. **The WMPU Dev blog.** *How to Install XAMPP and WordPress Locally on Windows PC.* <https://wpmudev.com/blog/setting-up-xampp/>